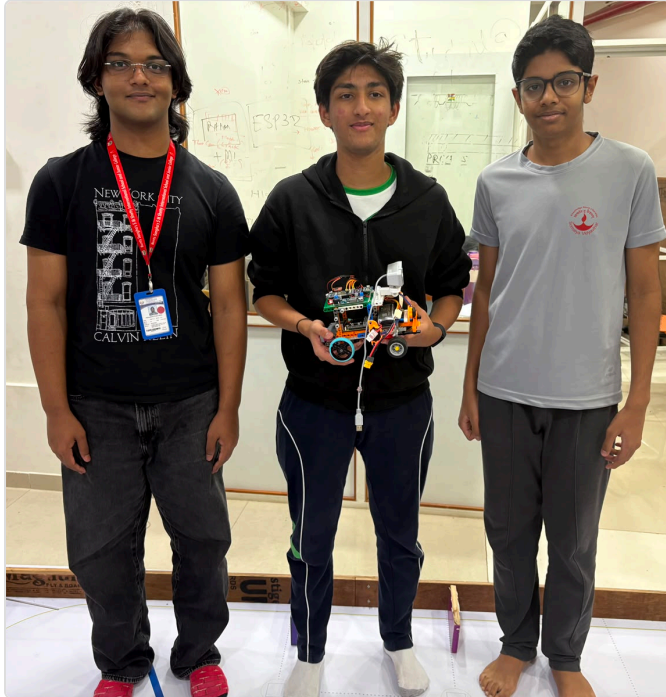


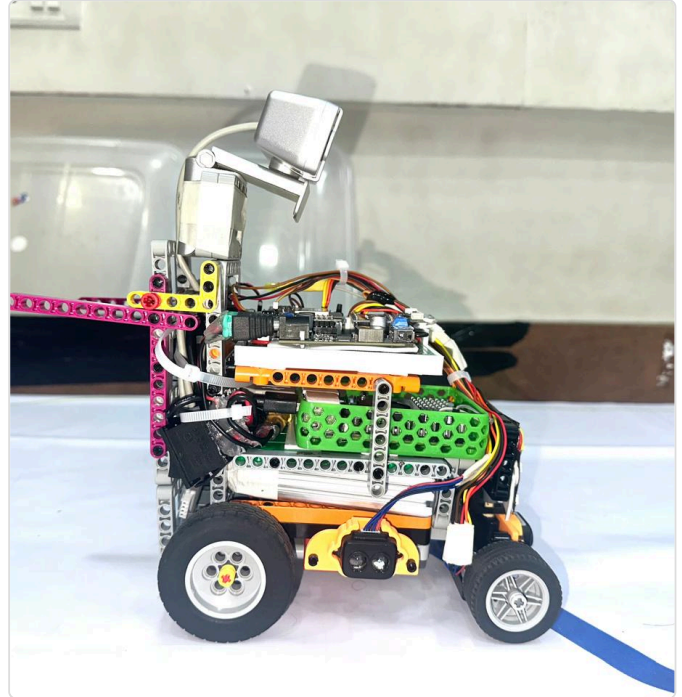
WRO FUTURE ENGINEERS 2026 · APAC

Engineering Journal

Team TED Drive · WRO Future Engineers APAC 2026



The team



The vehicle

Team	Ethan Fernandes (software) · Tejas Sirikonda (mechanical) · Deeyan Mehta (electronics)
Coach	Amey Chavan (OMOTEC)
Competition	WRO Future Engineers 2026, APAC, Hyderabad, September 2026
Repository	github.com/teddriveomo/wro-2026-future-engineers
Structure	WRO 2026 General Rules, Appendix C: sections 02 to 06 follow its five criteria

The repository holds the code, the measurements, the car's run logs and the documents this journal is built from, so the two always say the same thing.

01 Snapshot

The APAC vehicle in one page; every figure is from this journal and the repository.

Vehicle	An 84-step LEGO Technic frame (332 g bare), a steering column that turns the whole front axle 60° each way, the camera on a 24.8 cm tower pointing 12° down, six WS2812 LEDs on the front frame.
Mass	862 g race-ready with the Raspberry Pi 5 (measured 16 September).
Drive	MEX 6 V motor through a TB6612FNG driver to the rear wheels: 422 rpm free-running at the wheel and 0.89 m/s average over 3 m from a standing start (measured 21 September).
Power	3S 2200 mAh pack: 1.34 A idle, 1.93-2.02 A driving, 2.77 A worst case (each branch measured, 21 September).
Compute	Raspberry Pi 5 (camera, detection, decisions) and ESP32 (drive motor, servo, three TF-Luna, LEDs), linked by USB serial.
Software	<code>main.py</code> : Lab-colour pillar detection with tape-line and black-wall corner cues, a state machine that recentres between the walls after every pillar and corner and steers off a side wall during pillar passes, two pillars per straight, and a hand-over to a parking module after the twelfth corner (§04).
Open Challenge	<code>open-challenge.ino</code> on the ESP32 alone: direction detection, wall following at 30 cm with heading hold, twelve 90° turns on the BNO055 heading, and a stop 150 cm from the wall ahead (§04).
Evidence	The car's own logs: 27 runs from the parking lot reached the twelfth corner, the fastest in 154 s; <code>parallel_parking.py</code> parked in 15 of 32 attempts (§07). Bench measurements of drive, power and camera geometry (§02, §03); detection shown on the mat (§04); code stored byte-for-byte with MD5 checksums (§06).

Contents

§	Section	Appendix C
01	Snapshot	
02	Mobility: frame, drive, steering, camera placement, six views	C1
03	Power and sensing: LEDs, measured power budget, failure points, ESP32 pins	C2
04	Software and obstacle strategy: control loop, states, recovery, detection, parking, validation	C3
05	Engineering decisions: every change with its reason, evidence and cost	C4
06	Reproducibility: files, serial link, settings, provenance and checksums	C5
07	Validation: the car's run logs, 15-22 September	C3, C4

02 Mobility

How the car is built and moves (Appendix C, criterion 1).

Scope. The vehicle that competes at WRO Future Engineers APAC 2026, built after the India National Championship (26–28 Aug 2026): what changed and why. Every figure is marked by its basis: *measured* on the vehicle, *team* figure not yet measured, *derived* by calculation from named inputs, or *vendor* specification. Documents 1–5 remain the record of the Nationals configuration.

1. Specification

	Nationals (Aug 2026)	APAC 2026 (Sep 2026)	Basis
Length × width × height	length 23.0 cm	19.1 × 14.3 × 27.2 cm (±0.5 cm)	measured
Mass	785 g	862 g race-ready, Raspberry Pi 5 included (measured 2026-09-16)	measured
Axle loads	—	front 201 g, rear 661 g (23 % / 77 %) — see §2	measured
Chassis	LEGO Technic space-frame, 69-step build	LEGO Technic frame, 84-step build (instructions); 332 g bare	build file; measured
Printed parts	TF-Luna holders, servo horn beam, motor clamp (models/)	the same STL files	team
Steering	single-servo Ackermann (MG90S)	non-Ackermann: an MG90 servo drives a vertical steering column that turns the front axle, 60° each side	team
Wheelbase	—	13 cm	team
Wheels	front 41 mm, rear 55.6 mm	front 45 mm, rear 55 mm (rear tyres chosen for grip)	team
Drive motor	N20 gearmotor, 12 V, 600 rpm, via TB6612FNG	MEX motor (Avishkaar), rated 6 V, 400 rpm, via TB6612FNG — see §3	team
Gearing	20T pinion → 28T crown on a LEGO differential (5:7)	unchanged	team
Battery	3S LiPo, 11.1 V, 2200 mAh	unchanged	team
Power	pack → fast-charge module → Pi 5; pack → Buck-2 → 5 V rail (TF-Lunas, servo); TB6612FNG from the pack (2 — Power & Sensors)	unchanged, plus the WS2812 LEDs on the 5 V rail; power switch on the battery lead; start button on the rear beams	team
Sensors	3 × TF-Luna on a PCA9548A, BNO055, USB camera on a front arm at ~10 cm	same sensors; the camera (Lenovo 300 FHD) sits on top of a rear tower — see §4	photos; measured
Lighting	none	6 × WS2812 LEDs (three 2-LED strips) on the front frame — see §5	team

Wiring: [schemes/wiring_apac_2026.md](#). Photos: [v-photos/apac-2026-09-22/](#) — six views, 2026-09-22.

2. Weight and balance

Quantity	Value	Basis
Total, race-ready	862 g	measured
Front / rear axle	201 g / 661 g (23 % / 77 %); the two sum to the total	measured
Centre of mass	10.0 cm behind the front axle (13 cm × 661 / 862)	derived
Bare chassis	332 g, split 31 % / 69 %	measured
Everything added to the chassis	530 g, about 82 % on the rear axle	derived

Assuming the added parts share the chassis's split would have put 67 g too much on the front, which is why the full car was weighed per axle.

The driven rear wheels carry 77 % of the weight, which helps traction. The steered front wheels carry 23 %, on the smaller tyres, so front grip is what limits cornering. Planned check: measure the turning circle at walking pace and at race speed; a much wider circle at speed means the front is sliding.

3. Drive

- **Motor:** MEX motor (Avishkaar), rated 6 V and 400 rpm no-load (team).
- **Speed at the rated voltage:** $400 \text{ rpm} \times 20/28 = 286 \text{ rpm}$ at the rear wheels; $\times \pi \times 55 \text{ mm} = \mathbf{0.82 \text{ m/s no-load}}$ (derived).
- **Voltage the motor sees:** the TB6612FNG switches the pack (11.1 V nominal, 12.6 V full) with PWM, so the motor's average voltage is duty $\times$ pack voltage. With the Nationals firmware, full commanded speed was 100 % duty (12.47 V measured at the motor terminals) and half speed was 48 % duty (5.95 V) — see 2a. On a 6 V motor, duty above about 48 % on a full pack averages more than its rating. The APAC firmware writes the duty directly with an 8-bit `analogWrite` and no cap: `SPEED = 210` is 82 % duty, 9.1 V average on the nominal pack and 10.4 V on a full one, 1.5–1.7 times the motor's 6 V rating (derived). Turning (100, 39 %), parking (70, 27 %) and the collision escape (80, 31 %) stay under the rating. The motor temperature after a full 3-minute run has not been measured yet.
- **Measured 2026-09-21** (pack at 12.3 V, `SPEED = 210`, so 10.1 V average; ammeter in the motor wire unless stated):

Test	Result	Basis
Rear wheels lifted, free-running	0.33 A; rear wheel 422 rpm (slow-motion video of a taped wheel, counted over one minute) = 1.22 m/s	measured
Driving on the mat	0.599 A	measured
Wheels held (stall), ammeter on the battery lead	1.24 A, which is 1.51 A through the motor at 82 % duty	measured; motor figure derived
3 m straight line from a standing start	3.36 s, 0.89 m/s on average with the launch included	measured

- **Speed check.** Scaling the rated 400 rpm linearly to 10.1 V predicts 482 rpm at the rear wheel. The measured 422 rpm is 12.5 % lower, because linear scaling ignores the voltage lost in the winding and the drivetrain friction that shows up as the 0.33 A free-running current.
- **Motor model** (derived). Treating the motor as a standard DC motor (voltage = current $\times$ resistance + speed $\times$ motor constant) and fitting it to the free-running and stall points gives 6.71Ω including the driver and $0.128 \text{ V}\cdot\text{s}/\text{rad}$ at the gearmotor output. At the measured 0.599 A on the mat it predicts 0.94 m/s, and 3.37 s for the 3 m standing start (launch time constant 0.17 s). The measured time is 3.36 s.
- **Torque and grip** (derived; drivetrain efficiency beyond the free-running losses assumed 0.8). The rear axle gets about $0.04 \text{ N}\cdot\text{m}$ while cruising (1.4 N at the tyres, the rolling and drivetrain load on the mat) and $0.17 \text{ N}\cdot\text{m}$ at stall (6.2 N at the tyres). The rear tyres carry 6.5 N (661 g), so 6.2 N would need a friction coefficient of 0.95: a hard launch or a push against a wall spins the rear wheels on most surfaces before the motor stalls.

4. Camera placement

- **Mount:** lens centre $24.8 \pm 0.5 \text{ cm}$ above the floor; optical axis 12° below horizontal (78° from straight down); the hinge is glued at that angle (measured).
- **Field of view:** Lenovo lists a 95° diagonal field of view ([product page](#)), which is about 56° vertical and 87° horizontal at 16:9 (derived; not measured on this camera).

Quantity	Formula	Value
Where the optical axis meets the floor	$h / \tan \alpha$	117 cm ahead of the lens; about $\pm 10 \text{ cm}$ per 1° of angle error
Nearest floor in view	$h / \tan(\alpha + \text{VFOV}/2)$	$\sim 29 \text{ cm}$ from the lens; the lens sits $\sim 10 \text{ cm}$ behind the front wheels, so $\sim 19 \text{ cm}$ ahead of them
Separation of pillar bases at 0.5 m and 1.0 m	$\text{atan}(h / 0.5 \text{ m}) - \text{atan}(h / 1.0 \text{ m})$	12.5° (5.6° with the Nationals 10 cm mount)

Quantity	Formula	Value
Tops of walls 1–3 m away	$\alpha - \arctan((h - 10 \text{ cm}) / D)$	3.6°–9.2° above the image centre

h = lens height, α = pitch below horizontal, VFOV = vertical field of view, D = distance to the wall.

Why the camera went up. Reading range from the image depends on how low a pillar's base sits (the Nationals rule, [2 — Power & Sensors §3](#)). Raising the lens from ~10 cm to 24.8 cm makes that signal about 2.2 times stronger (12.5° against 5.6° between pillars at 0.5 m and 1.0 m), and the centre of the image lands ~1.2 m ahead instead of 0.33–0.57 m. The separation keeps growing with height up to $h = \sqrt{(Z_{\text{near}} \cdot Z_{\text{far}})}$, 55 cm for pillars at 0.3 m and 1.0 m, so the earlier statement in §3 of that document, that a higher mount compresses it, was wrong; it is corrected there.

What it costs.

1. A blind zone: a pillar within ~19 cm of the front wheels has its base below the frame, so the nearest-pillar rule needs a tie-break for boxes cut off at the bottom edge.
2. A view over the 100 mm walls: off-field objects can appear between the wall tops and the horizon, so the pillar search has to stay below the black wall band, not a fixed image row.

Planned checks: a tape on the floor slid until it just enters the bottom of the image (measures the blind zone directly); a start-position reference frame compared before each round (the hinge is glued, but this catches knocks).

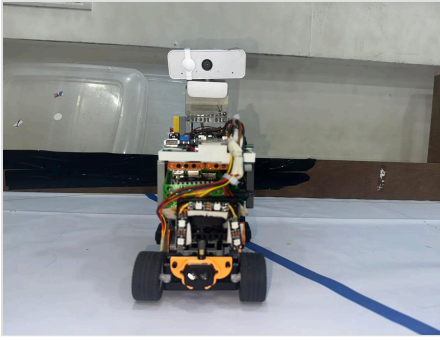
Colour calibration

The colour thresholds are set on the competition mat during one practice round at each venue, under that venue's lighting (team). Two tools in [src/apac-2026/tools/](#) apply the same image processing as `main.py`: a contrast and brightness scale, a gamma curve, then contrast-limited histogram equalisation (CLAHE, clip limit 2.0, 8 × 8 tiles) on the lightness channel.

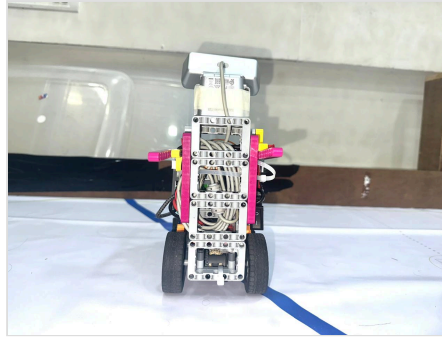
1. **Image.** `camera-settings.py` shows the live camera image with brightness, contrast and gamma sliders. The values chosen go into `BRIGHTNESS`, `CONTRAST` and `GAMMA` in `main.py` (committed: 0, 3.0 and 0.7).
2. **Colours.** `lab-calibration.py` converts the processed image to Lab and shows a mask while the lower and upper L, a and b sliders move. For each colour the sliders are set until only that colour shows in the mask, and the six values go into `COLOR_RANGES` in `main.py`.

Lab keeps lightness (L) apart from the two colour axes, so red (high a) and green (low a) can be separated with L left fully open, and a brighter or dimmer hall mostly changes L. The committed ranges do this: red is $a \geq 184$ and $b \geq 108$, green is $a \leq 101$, both with L from 0 to 255. The blue and orange tape lines are found with fixed HSV ranges (`COURSE_LINE_HSV_RANGES` in `main.py`), not with these tools.

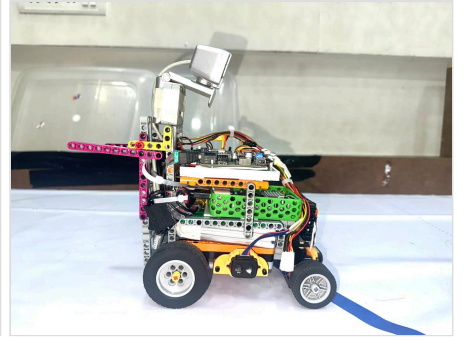
The six views



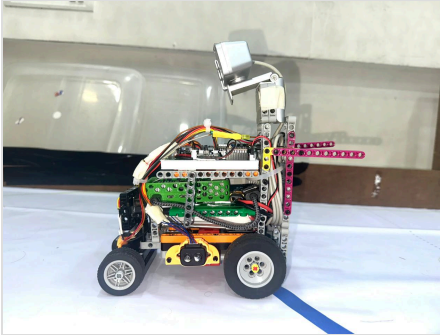
Front



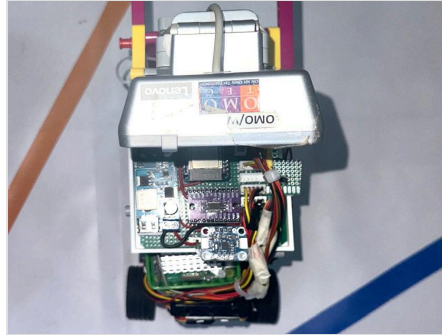
Rear



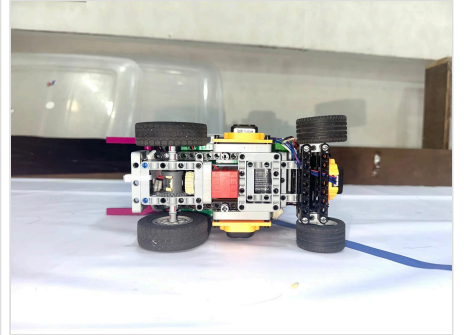
Left



Right



Top



Bottom (laid on its side)

03 Power and sensing

What powers the car, what it senses with, and what can fail (criterion 2).

5. Lighting and power additions

- **LEDs:** 6 × WS2812 on the front frame; data from ESP32 GPIO4; power from the 5 V Buck-2 rail (team). The firmware lights them white at `setBrightness(80)`, about 31 % (the bench measurements were taken at 50, about 20 %): roughly 19 mA per LED and 0.11 A for all six by the 60 mA-per-LED full-white rule of thumb in the [Adafruit NeoPixel Überguide](#) (derived). Full white would be 0.36 A.
- **Switch and button:** one power switch on the battery lead; the start button on the rear beams (team). The rules require exactly one power switch and one start button ([2 — Power & Sensors §6](#)).

Power budget, measured 2026-09-21

Each branch was measured on its own lead from the pack (ammeter and voltmeter, pack at 12.3 V). Pack currents for the two 5 V converters are given at the nominal 11.1 V, the conservative case used in [2a](#), with the converters assumed 88 % efficient; the motor branch is as measured.

Branch	Feeds	Basis	Idle	Driving	Worst case
Buck-2 (5 V)	3 TF-Luna, servo, 6 LEDs	measured at its input	0.13 A	0.23–0.32 A (servo sweeping)	0.32 A
Fast-charge module (5 V)	Pi 5, camera, ESP32	measured peak, 11.8 W at the Pi	1.21 A	1.21 A	1.21 A
TB6612FNG	drive motor	measured	0	0.49 A (0.599 A × 82 % duty)	1.24 A (stall)
Pack total			1.34 A	1.93–2.02 A	2.77 A

With the Pi's peak treated as its average, a 2200 mAh pack used to 80 % gives about 52 minutes of driving, about 17 three-minute runs (derived). The prediction for the Nationals car in [2a](#) was 0.95–1.40 A driving; the difference is the Pi (an 11.8 W peak against 6.2–8.5 W predicted as an average) and the larger motor.

Part	Load	Rating (vendor)	Use
TB6612FNG, one channel	1.51 A through the bridge at stall; 0.60 A driving	1.2 A continuous, 3.2 A peak	126 % at stall ; 50 % driving
Pi 5 USB ports without USB-PD	camera + ESP32	600 mA in total	see §6
Battery, 3S 2200 mAh 60C	2.77 A worst case	132 A	2 %

6. New failure points

Risk	Effect	Check or mitigation
Pi 5 fed from a USB-A fast-charge module (no USB-PD)	The Pi peaked at 11.8 W (2.36 A at 5 V, measured) and the module's rating has not been read; without USB-PD the Pi limits its USB ports to 600 mA in total, and the camera and the ESP32 both draw from them	Run <code>vcgencmd get_throttled</code> after a full run; move to a 5 A USB-C PD supply if it reports problems
WS2812 data at the ESP32's 3.3 V	The LED input-high threshold is $0.7 \times V_{DD}$ (3.5 V at 5 V); a marginal signal can flicker or show the wrong colour, tinting the scene	Watch for flicker in the camera image; add a level shifter if seen
6 V motor on the 11.1 V pack	Heat and brush wear: <code>SPEED = 215</code> averages 9.4–10.6 V, above the 6 V rating (§3)	Temperature check after a 3-minute run

Risk	Effect	Check or mitigation
Drive motor held against a wall	1.51 A through the TB6612FNG at stall, above its 1.2 A continuous rating (§5); the driver's thermal shutdown or the motor's heating would stop the car	The Pi's distance logic backs away (escape at 31 % duty for 0.30 s, at least 3 cm of progress, at most 2 attempts); on most surfaces the rear wheels spin before a full stall (§3)
Firmware front stop: the ESP32 stopped the drive motor when the front TF-Luna read under 18 cm or its reading went stale	Observed: it stopped the car and ended a run (team)	Removed from the firmware on 2026-09-10; collision handling moved to the Pi (§7)
Camera angle	Pixel-based settings shift if the angle moves	Hinge glued; reference frame before each round
Background above the walls	Off-field red or green objects read as pillars	Search only below the wall band
Blind zone ahead of the wheels	The nearest-pillar rule ties for boxes cut off at the bottom	Tie-break rule in the code

8. ESP32 pin map

From the Nationals firmware ([round2_ino.ino](#)) and the 2026-09-20 wiring draft; only GPIO4 is new. Checked against the APAC firmware ([src/apac-2026/](#)) on 2026-09-21: it reads the centre TF-Luna on mux channel 3 (channel 1 at Nationals).

Signal	GPIO
I ² C SDA / SCL — PCA9548A at 0x70: ch0 left, ch3 centre, ch2 right TF-Luna (0x10), ch4 BNO055 (0x28)	21 / 22
Steering servo	13
TB6612FNG AIN1 / AIN2 / PWMA / STBY	25 / 26 / 33 / 27
WS2812 data	4
Start button	32

04 Software and obstacle strategy

How main.py drives the Obstacle Challenge and how the car parks (criterion 3).

Scope. How the Obstacle Challenge code in `main.py` works: the control loop and the run (§1–§6), validation (§7), the parking modules (§8), pillar and line detection (§9), staying off the walls (§10) and the Open Challenge firmware (§11). Names in capitals are states; names in `code` are constants, given with their values.

1. The control loop

In a loop that sleeps 10 ms between passes (`NAVIGATION_LOOP_SECONDS`), the Raspberry Pi:

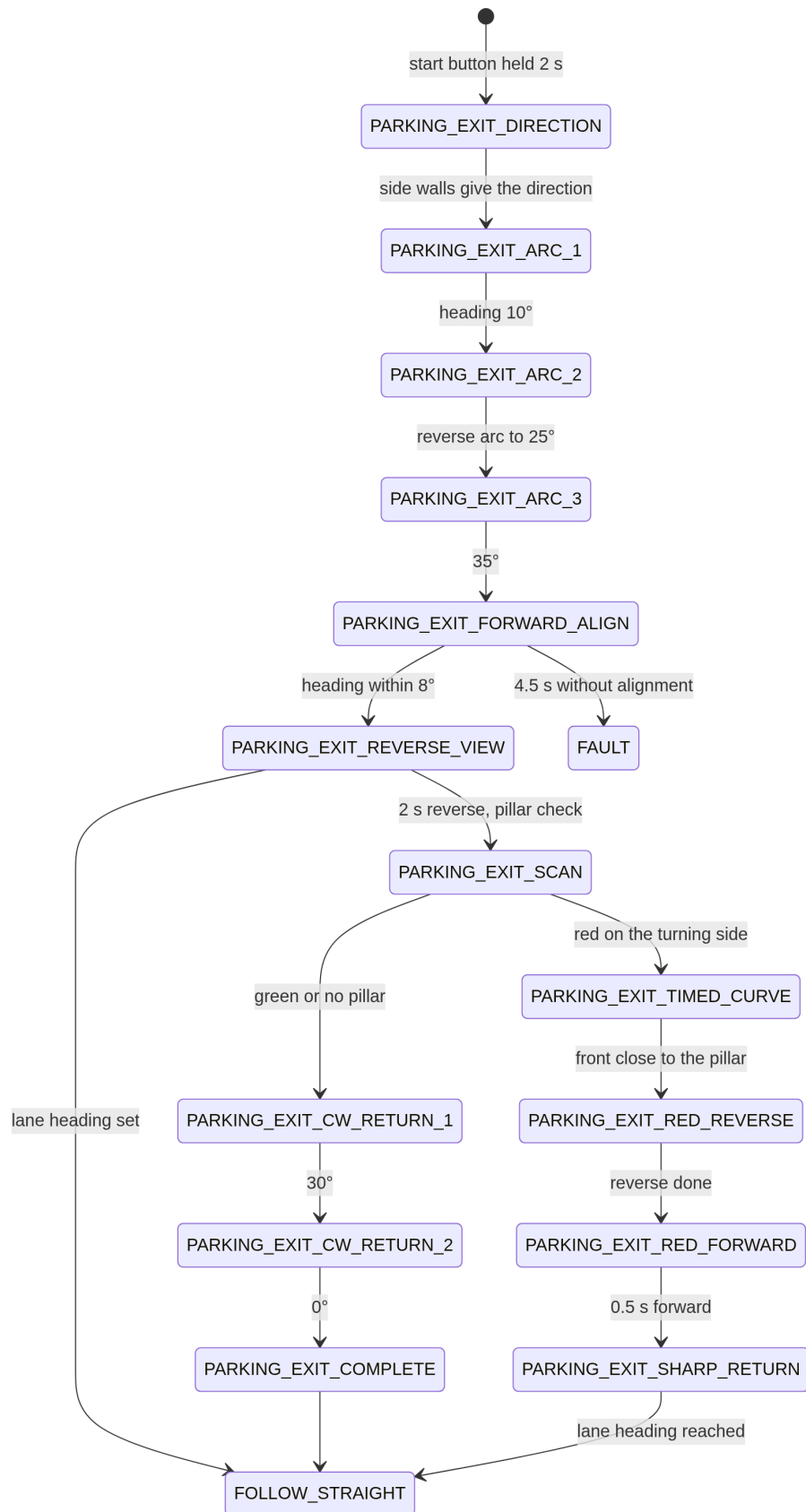
1. reads the newest ESP32 telemetry line: heading, the left, centre and right TF-Luna distances, and a validity mask;
2. takes the newest camera frame, masks its sides and runs the pillar and tape-line detector (`UnifiedPillarDetector`, §9);
3. calls the controller (`ObstacleChallengeController.update`), which sends one `speed,direction,steer` command to the ESP32.

The run starts once the start button has been held for 2 s: the ESP32 then sends `OK` and the Pi answers `OK`. The car leaves the parking lot (`ENABLE_PARKING_EXIT = True`), drives three laps and, after the twelfth corner (`COUNTER_MAX = 12`), hands over to a parking module (`PARKING_PARALLEL_HANDOFF`, §8). Every state change is printed as `[NAV] OLD -> NEW: reason`, and logged when `ENABLE_RUN_CSV_LOGGING = True` (§7).

2. The run

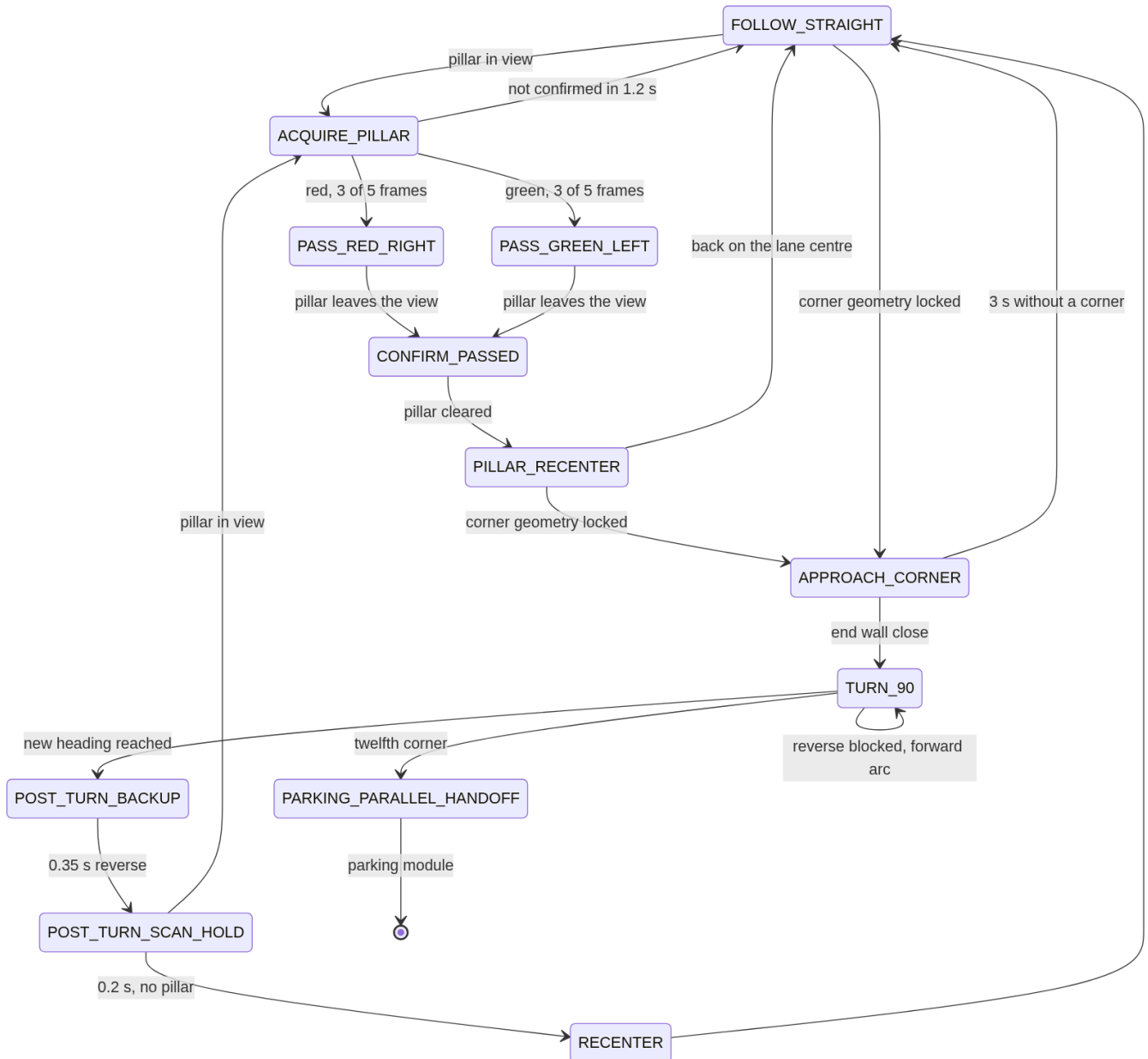
The diagrams in this section and in §3 show the transitions the car took in its logged runs of 19–22 September (89 runs, §7); the labels paraphrase the reasons the code prints.

Leaving the parking lot.



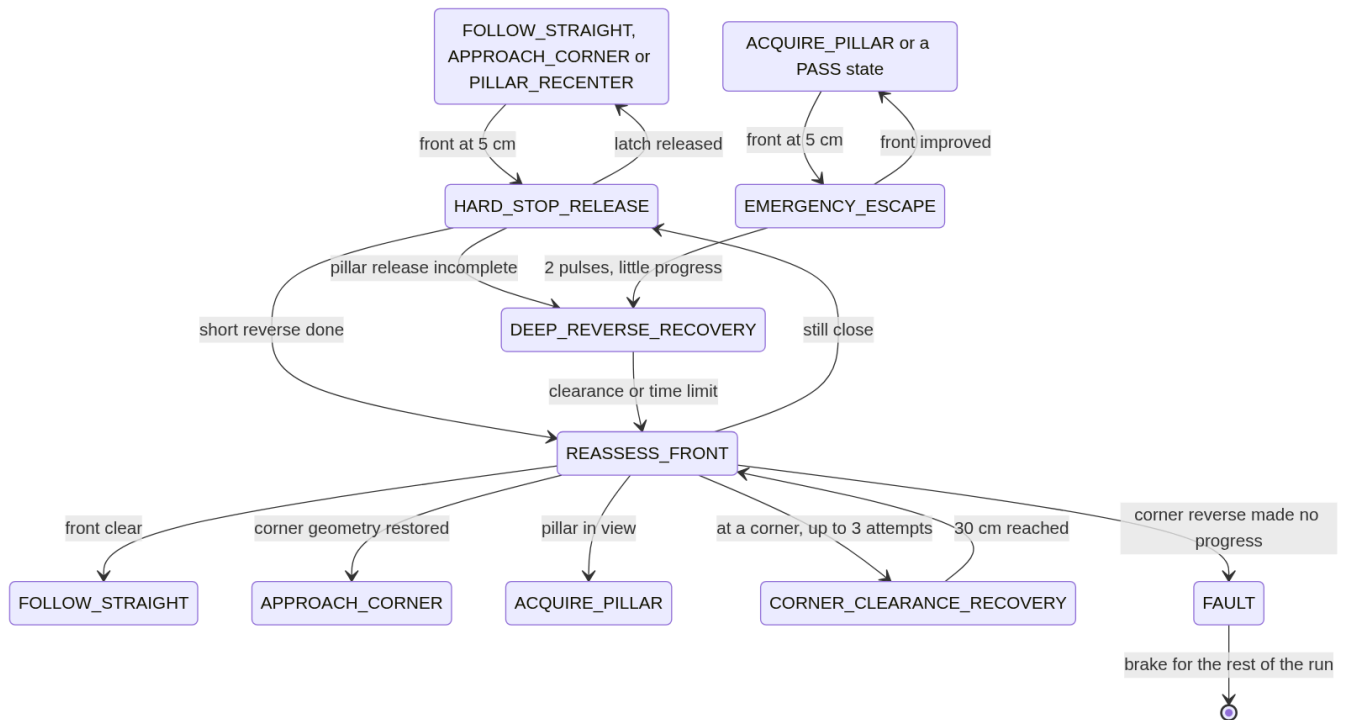
An anticlockwise start mirrors the lane return (PARKING_EXIT_ANTI_RETURN_1 and _2).

Three laps.



- Red pillars are passed on the right and green ones on the left. If the pillar is still in view after CONFIRM_PASSED, the car returns to its pass state.
- At most two pillars are taken on each straight (`MAX_PILLARS_PER_STRAIGHT = 2`); a pillar seen beyond both tape lines within 5 s of a pass (`ALPHA_POSITION_WINDOW_SECONDS`) belongs to the next straight, so the corner is taken first.
- Every driving state brakes and holds while the telemetry is older than 0.30 s (`TELEMETRY_STALE_SECONDS`), a TF-Luna or the heading is invalid, or the camera frame is older than 0.30 s (`CAMERA_STALE_SECONDS`). It resumes in the same state, and the hold does not count against that state's timers.

3. Recovery and faults



EMERGENCY_ESCAPE reverses for 0.3 s at speed 80 and resumes the state it interrupted once the front has improved by at least 3 cm (EMERGENCY_ESCAPE_*); after 2 pulses without that progress, DEEP_REVERSE_RECOVERY reverses at speed 75 for 0.6 to 0.8 s, aiming for 70 cm of clearance (DEEP_RECOVERY_*). HARD_STOP_RELEASE backs off until the front reads 30 cm (HARD_STOP_RELEASE_CM).

The code prints a reason with every FAULT: the turn could not reach its target after retries; the ESP32 did not report an applied reverse; the corner-clearance reverse made no reliable progress; both side walls were too close for a corner reverse; in the parking exit, the forward heading alignment timed out (4.5 s), the red approach took more than 4 s, or the timed return did not reach its heading within 5 s.

4. Why the states are split this way

State	Reason
PARKING_EXIT_DIRECTION	The lane direction must be known before the first corner. The code takes it from which side wall is nearer: 7 of the last 9 readings must agree, each with left and right at least 15 cm apart (DIRECTION_MIN_DIFFERENCE_CM).
PARKING_EXIT_ARC_1 to _3	Three heading arcs, the middle one in reverse, turn the car out of the lot to 35° without touching the parking-lot limitations.
PARKING_EXIT_FORWARD_ALIGN, PARKING_EXIT_REVERSE_VIEW	The car straightens to within 8° (at most 4.5 s), then reverses with the camera watching the start section, so the first pillar is seen before the car commits to a side.
PARKING_EXIT_SCAN	A pillar may stand just outside the parking lot. If it is on the side the car turns toward (red when driving clockwise, green anticlockwise), a timed pass keeps the car on its correct side; otherwise two heading arcs return the car to its lane.
ACQUIRE_PILLAR	One frame of red or green must not steer the car, so a pillar is followed only once 3 of 5 frames confirm it; an unconfirmed candidate is dropped after 1.2 s (PILLAR_ACQUIRE_TIMEOUT_SECONDS).
CONFIRM_PASSED	The camera loses a pillar before the rear of the car has cleared it, so the car holds its line for 0.25 s (PILLAR_CLEARANCE_HOLD_SECONDS).
PILLAR_RECENTER	A pass leaves the car angled towards a wall. A timed counter-steer alone would not prove the drift was cancelled, so the car counter-steers until its heading swings back past straight, then centres between the walls (§10).

State	Reason
APPROACH_CORNER	Neither a tape line nor a short front distance alone proves a corner. The approach locks after three fresh front readings of 105 cm or less; the turn needs the end wall within 35 cm (<code>OBSTACLE_CORNER_TRIGGER_CM</code>), the side the car turns toward open to at least 150 cm and the heading within 12° of the straight, in 3 agreeing samples over 0.25 s. After 3 s without a corner the car resumes the straight.
TURN_90	The target is the straight's heading plus or minus 90°, reached within 5° (<code>TURN_HEADING_TOLERANCE</code>). With at least 60 cm of room on the outer side (<code>CORNER_REVERSE_CLEARANCE_CM</code>) the turn starts with a reverse arc of up to 0.5 s; if a wall blocks the reverse, the turn continues as a forward arc.
POST_TURN_BACKUP, POST_TURN_SCAN_HOLD	After each corner the car reverses for 0.35 s at speed 125 and stops for 0.2 s, "so several stationary frames can prove a nearby pillar" before the next straight.
CORNER_CLEARANCE_RECOVERY	A car pressed against the end wall cannot turn. It reverses (speed 150, at most 0.4 s per attempt, up to 3 attempts) until the front has 30 cm of clearance, then re-checks the corner while stopped.

5. Edge cases the code handles

Situation	What happens
Telemetry or camera stale, or a sensor invalid	Brake and hold the current state; resume when fresh.
A third pillar on one straight	Ignored until the corner (live view: CORNER NEXT - NO NEW PILLAR).
Only one pillar on a straight	3.6 s after it, the second slot is filled and the corner takes priority (<code>SECOND_PILLAR_SLOT_TIMEOUT_SECONDS</code>).
Candidate pillar not confirmed	Back to FOLLOW_STRAIGHT after 1.2 s.
Pillar beyond the corner seen right after a pass	The corner is taken first (§2).
A side wall close during a pillar pass	A correction away from it from 18 cm, with fixed critical and emergency steers at 10 and 6 cm (§10).
Front at 5 cm	A short reverse, then a fresh check for a pillar, a corner or a clear front before moving on (§3).
Reverse blocked during a turn	The turn continues as a forward arc.
Both side walls too close for a corner reverse	FAULT (§3).
Twelfth corner	Hand-over to the parking module (§8).

6. Known limits

- The driving direction is locked at the start and does not change during the run.
- The last-resort front trigger (`PYTHON_EMERGENCY_RELEASE_CM = 5`) is inside the TF-Luna's blind zone: its specified range starts at 0.2 m (vendor). Corners are normally handled from 105 cm, so this trigger only acts after a misjudgement.
- The ESP32 always reports `hardStop = 0` (vehicle §7), so the `hard_stop` checks in `main.py` never fire.
- FAULT stops the car for the rest of the run.
- After the twelfth corner `main.py` hands over to the parking module its switches select; it stops with an error if both or neither are on (§8).
- Both parking modules end the entry when the front TF-Luna reads 5 cm or less (`ENTRY_STOP_FRONT_CM = 5`), and `partial_parking.py` also uses limits of 6, 8 and 15 cm (`REVERSE_BLOCK_SEARCH_SIDE_STOP_CM`, `REPOSITION_SIDE_CLEARANCE_CM`, `REPOSITION_FRONT_CLEARANCE_CM`). All are inside the TF-Luna's blind zone.
- In `parallel_parking.py`, `FINAL_REVERSE`, `FINAL_BRAKE` and `FINAL_FORWARD` are defined, but no transition leads into `FINAL_REVERSE`, so none of the three is reached; `PARALLEL_ALIGN` ends the parking in `COMPLETE`.
- The docstrings of `parallel_parking.py` say its detector does not use the blocks' colour; the code (`ParkingGeometryDetector.extract`) keeps only magenta pixels.

7. Validation

The metrics that exist, and how the next ones are collected.

- **Run results, 15–22 September** (the car's own logs, [APAC 2026 test logs](#)): 27 runs from the parking lot reached the twelfth corner, 22 of them on 18–20 September in a median of 176 s (fastest 154 s); `parallel_parking.py` parked in 15 of 32 attempts on 20 September.
- **Detector, per frame.** The live view shows the tracked pillar's confidence, hits out of 5, bottom position and area band (§9); in the two captured frames of 2026-09-12 the track had 5 of 5 hits at confidence 0.92 and 0.86.
- **APAC bench, 2026-09-21.** Drive and power measurements in [vehicle §3](#) and [§5](#): free-running and stall currents, 422 rpm at the rear wheel, 0.89 m/s over 3 m from a standing start, the current of each supply branch.
- **Run logs.** `main.py` can log every run: with `ENABLE_RUN_CSV_LOGGING = True` (committed as `False`) it writes `run_logs/round2_run_<date_time>.csv` beside itself, one row per telemetry sample and one per event, with lap, corner count, pillars this straight, state, heading and target, the three distances, the command sent and applied, the tracked pillar with its band, and the tape depths. Those files give corners completed, pillars passed, recoveries and faults per run; the 260 runs logged from 15 to 23 September are summarised in [APAC 2026 test logs](#). For parking, `parallel_parking.py` with `ENABLE_RUN_LOGGING = True` (committed as `False`) writes a CSV and an annotated video per attempt (§8).

8. Parking modules

Two files beside `main.py` park the car at the end of the run. After the twelfth corner `main.py` enters `PARKING_PARALLEL_HANDOFF` and calls the one its switches select (`PARTIAL_PARKING = True` and `PARALLEL_PARKING = False` as committed; exactly one must be on). Each can also run on its own, and each imports `main.py` for the camera, the serial link, telemetry, the live view and the shutdown. At the hand-over `main.py` first registers itself as the module `main (sys.modules)`, so the parking module reuses the camera, serial port and telemetry already running ("give the standalone parking module the same initialized camera, serial port, telemetry, and safety API"). Constants named in this section are in the parking files.

File	What it does	Start
parallel_parking.py	Parallel parking (<code>BUILD_ID = "slot-guided-entry-v29"</code>): finds the two parking-lot limitations with the camera, drives into the gap between them and straightens up	<code>python parallel_parking.py --direction clockwise --last-pillar auto</code>
partial_parking.py	Direct entry: the same approach and block search, then a drive into the space on a locked heading ("Standalone 90 degree parking" in its help text)	<code>python partial_parking.py --direction clockwise --last-pillar auto</code>

`--direction` is `clockwise` OR `anticlockwise`; `--last-pillar` is `auto` (detect a red or green final pillar), `red`, `green` or `none`. Both wait for the start signal as `main.py` does, and end in `COMPLETE` with the brake held (exit code 0) or in `FAULT` with a printed reason (exit code 1).

Which one. A full parallel park scores 15 points against 7 for a partial or non-parallel one (2026 General Rules, scoring element 1.8.2), and touching a limitation ends the round (rule 9.24.7). As committed, the switches select `partial_parking.py`; parking parallel means setting `PARALLEL_PARKING = True` and `PARTIAL_PARKING = False`. Every `main.py` name the parking modules use exists in the committed `main.py`.

Camera calibration. Both files need `parking_camera_calibration.json` beside them. It depends on how the camera is mounted; the team's file, made with the September camera mount, is committed beside them. To make a new one: `python parallel_parking.py --calibrate` shows the camera image, and clicking the four corners of a rectangle on the floor (near-left, near-right, far-right, far-left; 100 × 150 cm unless `--calibration-width-cm` and `--calibration-depth-cm` say otherwise) saves the homography from image to floor. If the file is missing, `parallel_parking.py` starts this calibration itself and exits after saving.

Finding the space (`parallel_parking.py`)

1. **Colour mask.** Pixels that are magenta in both HSV (`MAGENTA_HSV_LOW` 135, 65, 45 to `MAGENTA_HSV_HIGH` 179, 255, 255) and Lab ($a \geq 145$, $b \leq 175$). The top 28 % of the image is ignored (`ROI_TOP_FRACTION` = 0.28); one opening and two closings with a 5×5 kernel remove specks and fill gaps.
2. **Shape.** Each contour's rotated rectangle must fit a $200 \times 20 \times 100$ mm limitation seen face-on, edge-on or at an angle: aspect 1.15 to 6, rectangularity at least 0.42, magenta fill at least 0.45, width at most 30 % and height 5.5 to 72 % of the image, lower edge below 55 % of the image height and not cut off by the bottom of the image.
3. **Floor position.** The midpoint of the rectangle's two lowest corners goes through the calibration homography, which gives the block's position on the floor in cm.
4. **Two blocks.** A block is confirmed after 3 hits in its last 6 frames (`BOUNDARY_REQUIRED_HITS`, `BOUNDARY_HISTORY`). Block 1 is the best-scoring rectangle on the parking side of the image. Block 2 must not overlap it, and must be 25 to 120 cm from it on the floor, within 55 cm of it sideways and at least 15 cm farther away.
5. **Slot.** The slot is the floor midpoint of the two blocks. It is confident when both blocks are confirmed, neither has been missed for more than 2 frames, they are 25 to 120 cm apart and both have a confidence of at least 0.55. Its bearing from the camera steers the car (`SLOT_BEARING_KP` = 2.0) and is used only within $\pm 30^\circ$ (`SLOT_MAX_USABLE_BEARING_DEG`).

States (`parallel_parking.py`)

State	What the car does	Leaves when
SEARCH_LAST_PILLAR	Looks for the final pillar with <code>main.py</code> 's pillar detector	a pillar is confirmed: <code>PASS_LAST_PILLAR</code> ; no pillar and block 1 confirmed: <code>TRACK_OPENING</code>
PASS_LAST_PILLAR	Steers past the final pillar	red path: <code>SEARCH_FIRST_BOUNDARY</code> ; green path: <code>GREEN_RETURN_STRAIGHT</code> back to the start heading, <code>GREEN_FORWARD_STRAIGHT</code> for 1.0 s, then <code>SEARCH_FIRST_BOUNDARY</code>
SEARCH_FIRST_BOUNDARY	Drives at speed 50 on the start heading turned 45° towards the parking side (5° after a green-path pillar). If block 1 is not confirmed within 4 s, a short reverse (0.6 s at speed 60, steer 45°) and the search restarts; each retry adds 5° of offset (up to 55°), 0.2 s of search time and a longer, sharper reverse (up to 1.0 s, steer 55° ; the speed stays 60)	block 1 confirmed: <code>TRACK_OPENING</code>
TRACK_OPENING	The same search	block 2 confirmed and the slot confident: <code>ALIGN_VIRTUAL_CUTOUT</code>
ALIGN_VIRTUAL_CUTOUT	Speed 50, steering on the slot bearing (at most 25)	8 consecutive updates with a usable bearing (<code>ALIGN_CONFIRM_SAMPLES</code>): <code>ENTER_SPACE</code> ; front TF-Luna at 5 cm or less: <code>PARALLEL_ALIGN</code>
ENTER_SPACE	Speed 70, steering on the slot bearing (at most 12)	front TF-Luna at 5 cm or less: <code>PARALLEL_ALIGN</code>
PARALLEL_ALIGN	Speed 50 at full correction steer (50): reverse strokes of 0.8 s, then forward until the front TF-Luna reads 5 cm	heading within 2° of 0° on 3 consecutive updates: <code>COMPLETE</code>

The red and green paths swap in an anticlockwise run (`pillar_behavior`). After a green-path pillar, each block search ends in `FAULT` after 15 s (`GREEN_PARKING_SEARCH_MAX_SECONDS`), and the return to the start heading after 10 s.

What `partial_parking.py` changes

- **No final pillar.** With `--last-pillar none`, or when no final pillar is confirmed within 3 s, it drives straight for 2 s at speed 80 (`NO_PILLAR_STRAIGHT`), then searches for block 1 in reverse at speed 50 (`REVERSE_BLOCK_SEARCH`).
- **Green path.** Clockwise with a green final pillar (anticlockwise with a red one) it steers 45° forward until the centre TF-Luna reads 30 cm, then turns slowly at full lock the other way to a set heading before the block search

(GREEN_CW_TURN_RIGHT, GREEN_CW_TURN_LEFT; mirrored as RED_ACW_TURN_LEFT, RED_ACW_TURN_RIGHT).

- **Opening check.** Where the parallel module would start aligning, it requires a visible opening at least 6.5 % of the image wide (`MIN_VISIBLE_ENTRY_GAP_RATIO`) and a slot at least 20 cm ahead with a valid bearing. A narrower opening gets one repositioning (`REPOSITION_AWAY`, `REPOSITION_FORWARD`, `REPOSITION_RETURN`) before a `FAULT`.
- **Entry.** It locks the entry heading at the current heading plus the slot bearing (at most $\pm 30^\circ$) and drives in at speed 70, the parallel module's `ENTRY_SPEED` (`ENTER_SPACE`). Steering is $2.0 \times$ the heading error plus $0.6 \times$ its integral, which builds only once the slot is closer than 40 cm, plus a smoothed term of at most 8 that centres the opening in the image while the slot is 40 cm or more away; at most 24 in total.
- **Stop.** At a front reading of 5 cm or less, or when the front reading is missing or stale, it brakes (`FRONT_SENSOR_HOLD`): 3 fresh readings at the stop move on to `PARALLEL_ALIGN`, and 3 fresh clear readings resume the previous state. `PARALLEL_ALIGN` works as in the parallel module but counts only fresh telemetry.

Why it is built this way

Reasons as the code's comments give them.

Choice	Reason given in the code
The TF-Luna sensors are not used while searching for or aligning with the blocks	The blocks' geometry comes from the full camera view; distance matters only once the car is in the space
A short camera miss does not brake the car	The brake is not pulsed between camera frames: the selected motion continues until vision recovers, and during alignment the car drives straight and restarts the confirmation
Avoiding the final pillar starts on its first sighting, but the pass needs several frames of confirmation	At driving speed a one-frame delay can put the pillar past the useful steering window; a false detection can steer briefly but cannot latch the pass
The block search holds a heading offset towards the parking side	So that the blocks enter the camera view; an earlier path turned into the parking area before block 1 was confirmed
Reverse strokes in <code>PARALLEL_ALIGN</code> stop after 0.8 s; forward moves stop on the front TF-Luna	So that the car cannot back out of the space
The shape limits are broad	A limitation can be seen edge-on, face-on, at an angle or cut off by the image edge
One magenta contour never updates both blocks, and a sudden inward jump is taken as the next block	Block 1 leaves the image through its left edge in a clockwise run (the right edge anticlockwise), so a new rectangle further in is block 2, never a new sample of block 1
<code>partial_parking.py</code> ignores a pillar cut off by the image edge	Such an object belongs to the next lane, not to the final pillar ahead

Failure exits

Every `FAULT` prints its reason and holds the brake.

- `parallel_parking.py`: after a green-path pillar, the return to the start heading takes more than 10 s, or block 1 or block 2 is not found within 15 s.
- `partial_parking.py`: no valid IMU heading when the entry heading is locked; the opening is still narrower than 6.5 % of the image after one repositioning; the slot is not ahead with a valid bearing; something in front while the final pillar is classified, during the no-pillar straight or during the pillar pass; the pillar pass takes more than 4 s; too little side clearance in the reverse search; too little front or side clearance while repositioning; the opening is lost before the car passes it.

9. Pillar and line detection

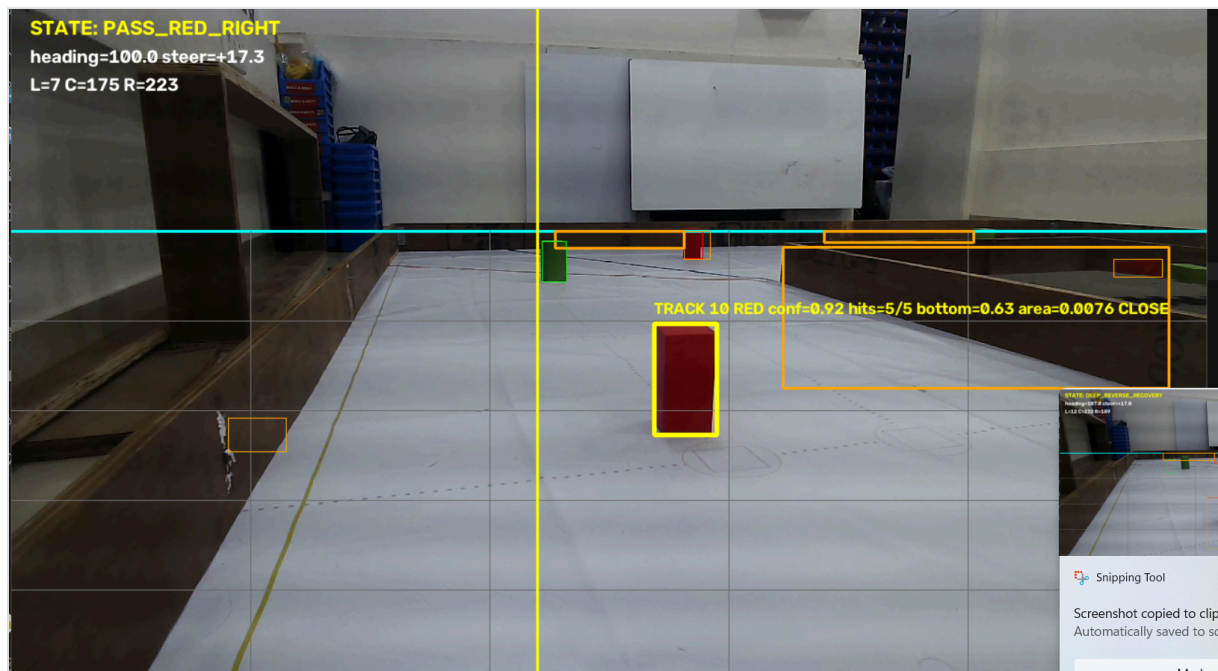
What `UnifiedPillarDetector` does with each 1280×720 frame, and why. Constants are in `main.py`; the two figures at the end show the result on the mat.

1. **Image preparation.** The frame is stretched (`CONTRAST = 3.0`, `BRIGHTNESS = 0`) and gamma-corrected (`GAMMA = 0.7`), then converted to Lab; CLAHE (clip 2.0, 8×8 tiles) evens out the lightness channel. Lab keeps lightness apart from the two colour axes, so red (high a) and green (low a) separate with L left open, and a brighter or darker hall mostly moves L ([vehicle §4](#)). The venue's thresholds come from one practice round with `lab-calibration.py`.
2. **Side masks.** Before detection a strip at each side of the frame is set to neutral grey (128, outside every Lab range) without moving pixel coordinates (`asymmetric_camera_view`): 16 % of the width on each side on a straight (`CAMERA_SIDE_MASK_FRACTION`). After a corner the two sides are masked separately, depending on the turn direction (`post_turn_view_fractions`, `POST_CORNER_VIEW_SIDE_FRACTION = 0.20`), until the new straight is established: 1.5 s of forward driving, or 12 s.
3. **Colour masks.** One Lab range per colour (`COLOR_RANGES`, as committed): red a at least 184 and b at least 108; green a at most 101; blue a 153 to 186 and b at most 90; orange a at least 130 and b at least 150; L unconstrained. Each mask is opened once and closed twice with a 3×3 kernel, and the top third of the image is cleared (`DETECTOR_ROI_TOP_RATIO`): from the 24.8 cm mount that third lies above the tops of the walls out to about 3 m ([vehicle §4](#)), so it holds only the hall.
4. **Candidate shapes.** Each contour is tested by size and shape. Pillars (red, green): area at least 0.045 % of the frame and height at least 2.5 % of the frame height; height to width 0.65 to 8 (a $50 \times 50 \times 100$ mm sign is about twice as tall as wide; the range allows partial views and perspective); rectangularity at least 0.35 and solidity at least 0.60, since a pillar is a solid rectangle and tape edges and reflections are not. Tape lines (blue, orange): area at least 0.03 %, height at least 1 %, height to width 0.2 to 10, rectangularity at least 0.25, solidity at least 0.50. Confidence = $0.20 \times \text{area} + 0.20 \times \text{height} + 0.25 \times \text{rectangularity} + 0.35 \times \text{solidity}$, with area and height capped at $5 \times$ the minimum area and 25 % of the frame height; candidates under 0.45 are dropped (`DETECTOR_MIN_CONFIDENCE`).
5. **Tape lines and the corner reference.** The mat's blue and orange lines mark the approach to a corner. They are found in HSV (blue hue 95-135, orange hue 3-22) below the top third, with the same morphology; a line must cover 0.025 % of the frame, span 10 % of its width and be at least twice as wide as tall. The lowest line's mid-height is its depth; when both colours are visible, the deeper of the two is the corner reference (the magenta line in the live view). The code also confirms a corner from the black walls: in HSV (value at most 70), a broad black end wall with an opening on the turn side, confirmed in 3 of the last 4 frames (`BLACK_CORNER_*`).
6. **Which pillar counts.** With the corner reference visible (and not suppressed for 3 s after a turn, `POST_TURN_TAPE_IGNORE_GRACE_SECONDS`), a pillar whose centre is above the reference, farther away than the tape, is ignored when it is entirely above it or sits in an outer column of the 5×5 grid: it stands beyond the corner or in a side lane (IGNORE CORNER in the live view). Two rules keep the car from dropping the obstacle in front of it: the track the controller is already passing keeps priority even when tape moves in front of it (`FRONT_PRIORITY`), and when a larger, lower interior pillar and a smaller outer pillar are both beyond the reference, the interior pillar stays actionable if it is $1.25 \times$ larger (`CORNER_FOREGROUND_AREA_ADVANTAGE`). A small ignored pillar never suppresses a valid foreground pillar.
7. **Tracking.** One pillar is tracked at a time. A new track starts on the best candidate by $0.55 \times \text{bottom} + 0.35 \times \text{confidence} + 0.10 \times \text{area}$: the lowest in the image is the nearest. Each frame the track is matched to a same-colour candidate by centre distance (at most 0.25 of the frame) or overlap (IoU at least 0.05), and its position is smoothed (0.65 new, 0.35 old). A track is confirmed after 3 hits in its last 5 frames (`DETECTOR_CONFIRMATION_HITS`, `DETECTOR_TRACK_HISTORY`); an unconfirmed track is dropped after 2 misses (`DETECTOR_UNCONFIRMED_MISSES`); a confirmed track stays locked until the controller records the pass, so a brief occlusion cannot switch the target. Confidence is $0.75 \times \text{the detection's confidence} + 0.25 \times \text{hits}/5$, and decays by 10 % per missed frame.
8. **Two pillars per straight.** At most two pillars are taken on each straight (`MAX_PILLARS_PER_STRAIGHT = 2`); once both are passed, no new pillar is taken until the corner (live view: `CORNER_NEXT - NO NEW PILLAR`). If no second pillar has come 3.6 s after the first, slot 2 is filled and the corner takes priority (`SECOND_PILLAR_SLOT_TIMEOUT_SECONDS`). One high track in the next lane is deferred rather than its whole colour suppressed, and right after a turn nearby pillars take priority (`defer_background_candidate`). A wide, low rectangle, at least twice as wide as tall, is treated as a parking-lot limitation, not a pillar; in laps 1 and 2 the car steers gently around it (speed 60, steering 15).
9. **Range and control.** The pillar's distance band comes from its area: far under $3,200 \text{ px} \times \text{px}$, near under 7,000, close under 20,000, otherwise too close; the 1280×720 breakpoints are kept as ratios of the frame (`PILLAR_AREA_NEAR_RATIO`, `PILLAR_AREA_CLOSE_RATIO`, `PILLAR_AREA_TOO_CLOSE_RATIO`). Each band selects a speed,

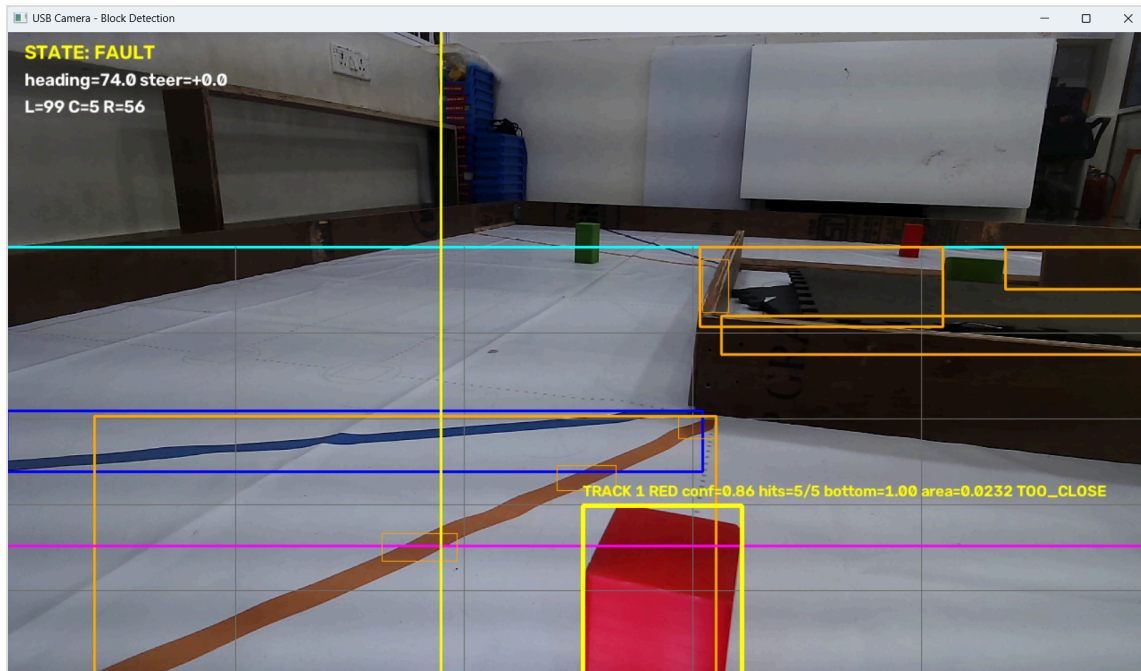
gain and steering limit (PILLAR_DISTANCE_PROFILES; far: speed 100, gain 0.70, 3 to 12 of steer; too close: speed 80, gain 1.50, 38 to 52). The controller steers the pillar to a target column, red to 38 % of the width so the car passes on its right and green to 62 % so it passes on its left (PILLAR_RED_TARGET_X, PILLAR_GREEN_TARGET_X), with gain 200 (PILLAR_LATERAL_KP), at most 40 of correction and smoothing 0.65.

The live view (annotate): the cyan line is the top of the search area; the grey grid is the 5×5 columns and rows; blue and orange boxes are tape lines; the magenta line is the corner reference; thin boxes are candidates; a magenta box is a pillar ignored as beyond the corner (IGNORE CORNER C_n, n = grid column); a yellow box marks the pillar being passed (FRONT PRIORITY); the tracked pillar's box is yellow once confirmed and cyan before, labelled TRACK id COLOUR conf hits/5 bottom area BAND, with a vertical line at its target column; the text lines give the state, heading and steer, and the three TF-Luna readings in cm. The live view also shows PILLAR SLOTS n/2, CORNER NEXT - NO NEW PILLAR, SLOT 2 TIMEOUT IN n s, BLACK CORNER and VIEW MASK L=n% R=n%.

Two frames from testing on 12 September.



Passing a red pillar: state PASS_RED_RIGHT, steer +17.3 at heading 100.0; TF-Luna L 7, C 175, R 223 cm; track 18, red, confidence 0.92, 5 of 5 hits, bottom 0.63, area 0.0076, band close; orange tape boxes ahead and the red target column drawn.



A run that ended in FAULT: a red pillar 5 cm from the centre TF-Luna (L 99, C 5, R 56 cm), track 1 at confidence 0.86, bottom 1.00, area 0.0232, band too close; both tape lines are visible and the magenta corner reference is drawn. The fault reason is printed to the terminal, not to the view, and was not captured.

10. Staying off the walls

Four mechanisms keep the car in the middle of its lane; with them the side rollers were no longer needed ([vehicle §7](#)).

- **PILLAR_RECENTER**, after every pillar pass, counter-steers 22° against the pass for 0.18 to 0.45 s and moves on once the heading has swung back past straight, "proof it actually cancelled the drift, not just a timer"; it then settles onto the lane centre from the heading and both side TF-Lunas (both walls 8 to 110 cm away, heading within 10° , gain 0.5, at most 18 of steer, 3 confirmations).
- **A wall guard during pillar passes**: a side wall closer than 18 cm (`PILLAR_WALL_WARNING_CM`) adds a correction away from it that grows to 35 at 6 cm; at 10 cm a fixed critical steer takes over, and at 6 cm an emergency steer at speed 65.
- **Recentering after corners**: a side-distance correction (`ROUND2_RECENTER_*`: gain 0.45, at most 14, heading within 8°).
- **CORNER_CLEARANCE_RECOVERY** (§4), with the side-clearance check before a corner reverse (`TURN_SIDE_CLEARANCE_CM = 20`).

11. Open Challenge firmware

[open-challenge/open-challenge.ino](#) drives the Open Challenge on the ESP32 alone, without the Raspberry Pi or the camera. It uses the same pins, I2C multiplexer channels and sensors as the Obstacle firmware; the servo centre is `SERVO_CENTER = 106` here and 85 in the Obstacle firmware. Constants are in the file.

1. **Start and direction.** After the start button (GPIO32), the car drives at speed 190 (`DETECTION_SPEED`) until one side reading opens beyond 100 cm on 3 samples (`DIRECTION_DETECTION_DISTANCE_CM`, `DIRECTION_CONFIRMATION_SAMPLES`); that side sets clockwise or anticlockwise, keeping the original priority for the right side.
2. **Wall following.** At speed 190 (`FOLLOW_SPEED`) the car holds its heading (gains 0.55 and 0.08, `HEADING_KP`, `HEADING_KD`) and keeps 30 cm from the followed wall (`TARGET_LEFT_DISTANCE_CM`, `TARGET_RIGHT_DISTANCE_CM`; gain 0.50, at most 12), with at most 25 of steering in total.
3. **Corners.** When the followed side opens beyond 75 cm on the left or 50 cm on the right on 3 samples, the car turns 90° on the BNO055 heading at speed 160 (gains 0.75 and 0.06, at most 35 of steering) until the error is under 4° . It then drives at speed 210 for a minimum time (500 ms clockwise, none anticlockwise) and reacquires the wall below 50 cm on 3 samples.

4. **Finish.** After twelve turns (`TOTAL_TURNS`) it holds the heading of the last turn and brakes when the centre TF-Luna reads 150 cm or less (`FINAL_CENTER_DISTANCE_CM`); the electrical brake then stays latched.

The control loop runs every 10 ms (`CONTROL_PERIOD_US`), and servo pulses come from a dedicated task on the other core (`servoPulseTask`, 20 ms frames).

05 Engineering decisions

Each change as a decision: what it replaced, why, the evidence, and the cost (criterion 4).

7. Why each change was made

Each change as a decision: what it replaced, the reason the team gives, the evidence in this repository, and what it costs. The largest test-driven change in this vehicle's line is in the Nationals record: the wheel-shed failure of 2026-08-06 led to the space-frame redesign that every later frame descends from ([1 - Mobility §3, D8](#)).

Change	Instead of	Reason (team)	Evidence here	Trade-off
New 84-step frame, 332 g bare	the Nationals 69-step space-frame	a sturdier structure	build file; the car weighs 862 g against 785 g at Nationals (measured, all changes together)	77 g more to accelerate and stop; stiffness not measured
Steering column turning the whole front axle, 60° each way	single-servo Ackermann	a smaller turning circle	turning circle not measured	the front wheels no longer follow Ackermann geometry, so they scrub in tight turns
Wheels: front 45 mm, rear 55 mm tyres	front 41 mm, rear 55.6 mm	rear tyres for grip	not measured	none recorded
Camera on a 24.8 cm tower, 12° down	a front arm at about 10 cm	a better view of the track	measured mount, derived geometry: pillar bases at 0.5 m and 1.0 m are 12.5° apart against 5.6° before, a 2.2 × stronger range signal (§4)	a 19 cm blind zone ahead of the wheels and a view over the walls (§4); a taller top, its centre-of-gravity height not measured
Six WS2812 LEDs on the front frame	ambient light	fewer colour mistakes	0.11 A at the committed brightness of 80 by the rule of thumb (0.07 A at 50, the bench setting); Buck-2 idle 0.13 A measured with them (§5); misread counts not measured	data at 3.3 V may flicker (§6)
Side rollers fitted, then removed	none	fitted to keep the car moving on wall contact; removed once the rewritten wall-avoidance code made them unnecessary, and because they blocked the Raspberry Pi's ports (reported 2026-09-22)	the team ran the car in both configurations without trouble (team); in the logs of 22 September evening, after the removal, 5 of 7 short test runs reached COMPLETE, and the corner-clearance recovery ran in 5 of them (test logs §4); the current <code>main.py</code> adds lane recentering after every pillar pass and after corners, a wall guard during pillar passes and a corner-clearance recovery (software §10)	without rollers a wall contact stops the car; only the Pi's collision escape (§6) remains
Firmware front stop removed (2026-09-10)	a stop under 18 cm on the front TF-Luna	stopping the drive motor ended a competition run	one run ended (team)	collisions are handled by the Pi's distance logic alone (§6)
Parallel parking at the end (<code>parallel_parking.py</code>)	direct entry (<code>partial_parallel_parking.py</code>)	15 points for a full parallel park against 7 (2026 rules, scoring element 1.8.2)	rules; logs: 15 of 32 <code>parallel_parking.py</code> attempts parked on 20 September, while the U-turn parking-in of 17 September reached COMPLETE in 0 of 24 (test logs §3)	a longer manoeuvre beside the limitations; touching one ends the round (rule 9.24.7)

The front-stop decision. The firmware used to stop the drive motor when the front TF-Luna read under 18 cm (released at 24 cm) or when its reading went stale, and that stop ended a run. On 2026-09-10

`forwardHardStopIsActive()` was changed to return `false`, with the comment "Disabled: stopping the drive motor here ends a competition run". Collisions are handled on the Pi: back away at speed 80 for 0.30 s, require at least 3 cm of progress, at most 2 attempts, release at 30 cm. The same day added a `PARKING_OVERRIDE_ON` serial command to the firmware; `main.py` never sends it, so it has no effect in a run.

06 Reproducibility

How to rebuild and run the software exactly as committed (criterion 5).

The code the vehicle runs at WRO Future Engineers APAC 2026: the Obstacle Challenge stack and the Open Challenge firmware. The vehicle is described in [docs/apac_2026_vehicle.md](#), and how `main.py` drives the run, with its state diagrams, in [docs/apac_2026_software.md](#). The two parking modules are described in its [section 8](#).

File	Runs on	What it does
obstacle-challenge/main.py	Raspberry Pi 5	Camera, pillar and line detection, and the driving decisions; sends drive commands to the ESP32
obstacle-challenge/parallel_parking.py	Raspberry Pi 5	Parallel parking at the end of the run (<code>BUILD_ID = "slot-guided-entry-v29"</code>): finds the two parking-lot limitations with the camera, drives into the gap and straightens up. Imports <code>main.py</code> ; called by <code>main.py</code> after the twelfth corner when <code>PARALLEL_PARKING = True</code> , or run on its own
obstacle-challenge/partial_parking.py	Raspberry Pi 5	Direct entry into the parking space on a locked heading, reusing the parallel module's approach and block search. Called by <code>main.py</code> after the twelfth corner when <code>PARTIAL_PARKING = True</code> (as committed), or run on its own
obstacle-challenge/esp32/obstacle-challenge-final-code/obstacle-challenge-final-code.ino	ESP32	Drive motor, steering servo, LEDs, the three TF-Luna sensors and the BNO055 through the PCA9548A; sends telemetry to the Pi
open-challenge/open-challenge.ino	ESP32	The Open Challenge on the ESP32 alone: direction detection, wall following with heading hold, twelve 90° turns on the BNO055 heading, and a final stop at 150 cm from the wall ahead (software §11)
obstacle-challenge/parking_camera_calibration.json	Raspberry Pi 5	The team's image-to-floor calibration for the parking modules (1280 × 720 image, 100 × 150 cm rectangle)
tools/lab-calibration.py	Pi or laptop	Interactive Lab colour-threshold picker (trackbars), with the same image preprocessing as <code>main.py</code> ; how it is used at a venue is in vehicle §4
tools/camera-settings.py	Pi or laptop	Interactive brightness, contrast and gamma preview

Serial link

115200 baud. `main.py` opens `/dev/ttyUSB0` on the Pi (`COM3` is its Windows bench setting).

- Pi to ESP32: `speed,direction,steer` and a newline. `speed` is the 8-bit PWM duty (0 to 255); `steer` is a correction of -60 to +60 around the servo centre of 85.
- ESP32 to Pi: telemetry lines `T,seq,ms,heading,left,center,right,validMask,hardStop,appliedSpeed`.

Settings as committed

Setting	Value	File
Drive speed	<code>SPEED = 215 (84 % duty)</code>	<code>main.py</code>
Turning, parking, collision escape	<code>100, 70, 80</code>	<code>main.py</code>
Leave the parking area at the start	<code>ENABLE_PARKING_EXIT = True</code>	<code>main.py</code>
Park at the end	<code>ENABLE_PARKING_IN = True</code> : after the twelfth corner <code>main.py</code> hands over to a parking module, chosen by <code>PARTIAL_PARKING = True</code> and <code>PARALLEL_PARKING = False</code> as committed (exactly one must be on; software §8)	<code>main.py</code>

Setting	Value	File
Run log	ENABLE_RUN_CSV_LOGGING = False; set to True, every run writes run_logs/round2_run_<date_time>.csv beside main.py, one row per telemetry sample and one per event	main.py
Parking modules	run on their own, e.g. python parallel_parking.py --direction clockwise --last-pillar auto	parallel_parking.py, partial_parking.py
Parking run log (CSV and video per run)	ENABLE_RUN_LOGGING = False	parallel_parking.py
Servo centre	SERVO_CENTER = 85 in the Obstacle firmware; 106 in the Open Challenge firmware	firmware
Mux channels	TF-Luna left 0, centre 3, right 2; BNO055 4	firmware
Front hard stop	disabled since 2026-09-10 (forwardHardStopIsActive() returns false); see vehicle §7	firmware

Both parking modules read parking_camera_calibration.json beside them. It depends on how the camera is mounted: the team's file, made on the practice mat with the September camera mount, is committed; python parallel_parking.py --calibrate makes a new one if the mount changes.

On the Pi, main.py sets GPIO3 high once it has initialised and is waiting for the start signal, and low when it exits.

Provenance and checksums

The files come from the team's working copy of the code (the coach's repository, commit 63c2b31, with the team's later changes) as of 23 September 2026. They are stored byte-for-byte (see .gitattributes), so these MD5s can be checked on any checkout:

File	MD5
main.py	77187817b99846d4c7d8766d5e9b8bf7
obstacle-challenge-final-code.ino	a0bf8c1ca5330442bed69d8299892572
lab-calibration.py	aad32eb3ca1954ad38683ee5a004da1a
camera-settings.py	f02e7f9eaae329e8867e965ac4f98511
parallel_parking.py	9d622df00da023df403fc06669c3c014
partial_parking.py	34bb36075c2ab680220bd6557056992a
open-challenge.ino	caf69d495f3d0f259a796dfcb81e5b5c
parking_camera_calibration.json	d2ebad50bf7a573496625ee7d20f26da

Not included, because none of them runs on this vehicle: the build without LEDs, the firmware and the Open Challenge variant for a smaller robot, backup copies, component tests, experiments (main_background_networking.py, corner_slot_filler.py, parking_entry_only.py, parking_exit_only.py, cornering_sequence.py) and the HSV and RGB colour tools.

07 Validation

What the car's own logs show, 15 to 22 September (criteria 3 and 4).

Scope. What the car's own logs record from 15 to 23 September 2026: 260 runs of `main.py` (`round2_run_*.csv`, one row per telemetry sample and one per event, written when `ENABLE_RUN_CSV_LOGGING = True`) and 32 attempts of `parallel_parking.py` (`parking_run_*.csv`, `ENABLE_RUN_LOGGING = True`). Every figure here is computed from those files by `run_metrics.py`. The per-run summary is `data/run_summary_2026-09.csv`, the parking attempts are in `data/parking_runs_2026-09-20/`, and `data/example_full_run_20260919_192712.csv` is the fastest full run, whole. The code was tuned from day to day, so each day's figures belong to that day's settings.

1. Full runs from the parking lot

A run counts here when it starts with the parking exit (first state `PARKING_EXIT_DIRECTION`). It reached the twelfth corner when `corner_counter` reached 12; the time is counted from the first drive command. Runs the team stopped by hand are included, so the counts are attempts, not a success rate.

Day	Runs	Reached the 12th corner	Time to the 12th corner, s: median (range)
15 Sep	22	1	241 (241–241)
16 Sep	53	2	225 (201–248)
17 Sep	15	2	206 (204–208)
18 Sep	39	9	180 (165–190)
19 Sep	51	10	167 (154–184)
20 Sep	7	3	176 (168–180)

From 15–17 to 18–20 September the time to the twelfth corner fell from 201–248 s to 154–190 s (median 176 s over 22 runs); 14 of those 22 runs reached the twelfth corner inside 180 s, the length of a round.

2. Why full runs stopped, 18–20 September

The code prints a reason with every `FAULT`. Reasons in the full runs of 18–20 September:

Reason	Runs
corner clearance reverse made no reliable progress	7
parking-exit forward heading alignment timeout	4
front reached 5 cm before parking alignment	1
unknown Round 2 state <code>STRAIGHT</code>	1
parking-exit reverse heading alignment timeout	1
red approach exceeded 4s without reaching distance trigger	1

The most frequent, the corner-clearance reverse that made no progress, is the stall of a car pressed against a wall at a corner.

3. Parking

U-turn parking-in in `main.py`, 17 September. 24 test runs of the U-turn routine; 0 reached `COMPLETE`. The faults were clearance limits during the U-turn (U-turn forward clearance only N cm: 5; U-turn side clearance only N cm: 3; U-turn heading did not complete; progress=N: 1). Earlier that evening, 10 runs of the timed routine (reverse at full left lock, then right) reached `COMPLETE` in 9; the log cannot show whether the car ended inside the space.

`parallel_parking.py`, 20 September. 32 attempts from 14:36 to 17:35; 15 reached `COMPLETE`, in a median of 31 s (20–88 s). In the last session, from 16:30, 7 of 11. The attempts that did not finish stopped in `TRACK_OPENING` (5),

ALIGN_VIRTUAL_CUTOUT (5), ENTER_SPACE (3), PARALLEL_ALIGN (2), PASS_LAST_PILLAR (1), FAULT (1).

Three laps, then parallel parking, 20 September. In 3 runs `main.py` handed over after the twelfth corner (PARKING_PARALLEL_HANDOFF) and `parallel_parking.py` took over; 2 of them ended parked:

Start	Time to the 12th corner, s	Parking	From the first drive command to the end, s
17:12	168	COMPLETE after 58 s	226
17:26	180	TRACK_OPENING after 13 s	193
17:32	176	COMPLETE after 88 s	264

At those speeds three laps plus parking took longer than a three-minute round.

4. After the side rollers came off, 22 September

7 short runs in the evening, each started on a straight (DETECT_DIRECTION) with a 2-corner limit, as the counters show; 5 reached COMPLETE. CORNER_CLEARANCE_RECOVERY ran in 5 of them, and in 5 the car went on to make its turn.

5. What the logs do not show

- Whether a pillar or a wall was touched: the logs record states, distances and commands, not contacts.
- How squarely the car ended in the parking space.

6. Performance videos

Challenge	Video	Run
Obstacle Challenge	video/apac-2026-obstacle-run.mp4 , 3 min 19 s	The APAC vehicle, about 20 September: from the parking lot, three laps (12 corners), then a stop after the twelfth corner
Open Challenge	youtu.be/PbOQJ54wr-0 , 33 s	India National Championship, August 2026 (the first vehicle)
Obstacle Challenge	youtu.be/dBjqMQ-jRBA , 100 s	India National Championship, August 2026 (the first vehicle)